



KDOM Project

PART A : Q3

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Given information :

The translating radial roller follower of a cam is to **rise 40 mm** with cycloidal motion in 180° of cam rotation and return with cycloidal motion in the remaining 180° . If the **roller radius is 10 mm** and the **prime circle radius is 50 mm**,

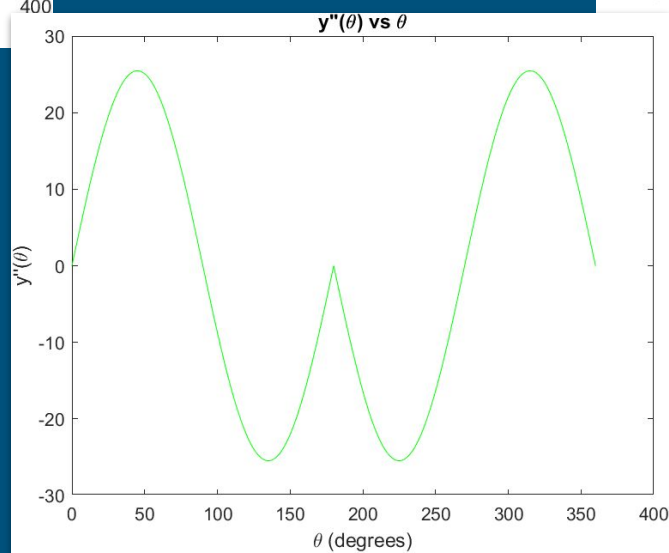
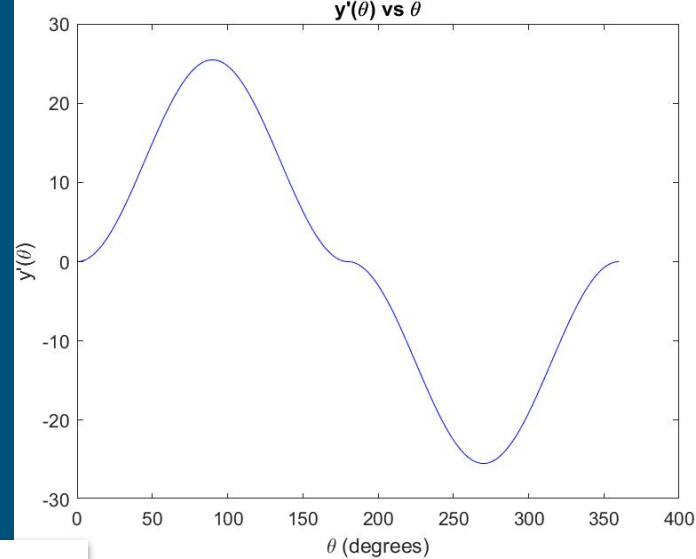
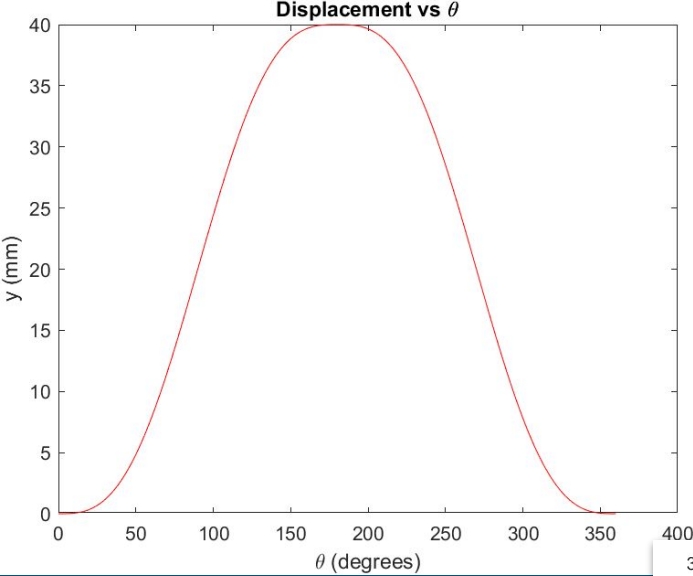
Formula for lift :

Since the motion is cycloidal, we have the lift as a function of theta given below:

$$y(\theta) = \begin{cases} L\left(\frac{\theta}{\theta_i} - \frac{\sin(2\pi\frac{\theta}{\theta_i})}{2\pi}\right) & \text{for } 0^\circ \leq \theta \leq 180^\circ \\ L\left(1 - \frac{\theta - \beta}{\theta_r} + \left(\frac{\sin(2\pi\frac{\theta - \beta}{\theta_r})}{2\pi}\right)\right) & \text{for } 180^\circ < \theta \leq 360^\circ \end{cases}$$

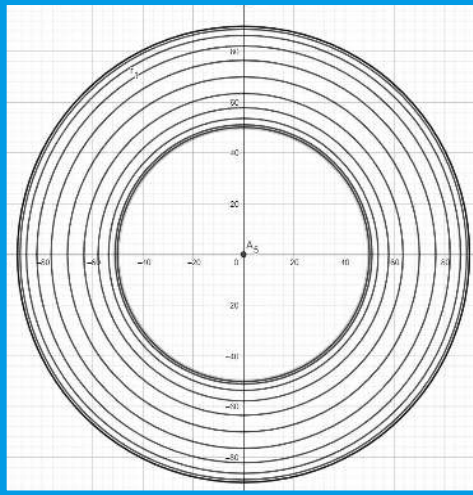
where,

$$\beta = 180^\circ, L = 40\text{mm}, \theta_r = 180^\circ$$



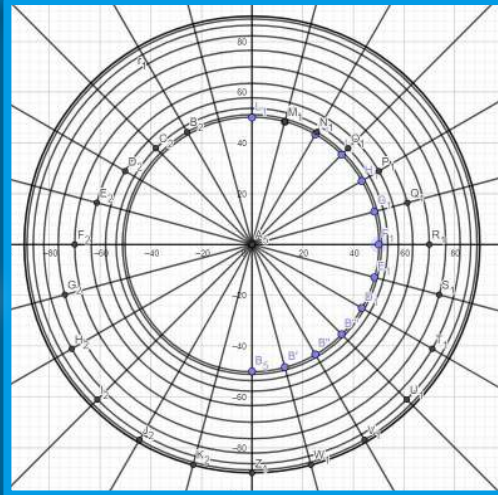
STEP1

Using the expression for lift as mentioned earlier, we plot circles with centre at origin and radii of base circle + value of lift expression at intervals of 15° , starting from 15° (15° to 180° i.e.).



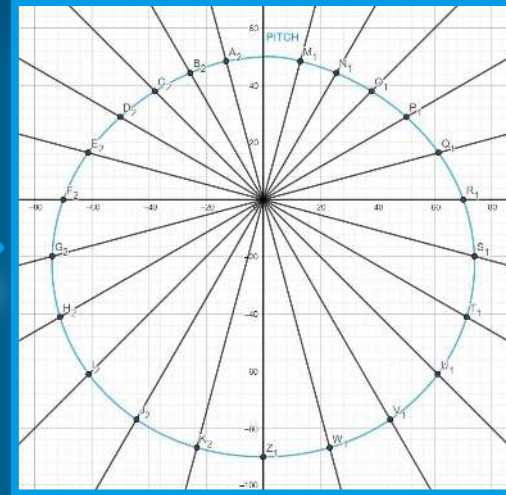
STEP2

Next, we take points 15° apart (from 0°) and draw lines through them from origin, which intersect the corresponding circle with the value of lift expression at that angle (e.g. for 15° , we intersect the circle of radius $50 + f(15^\circ)$) and we mark these points.



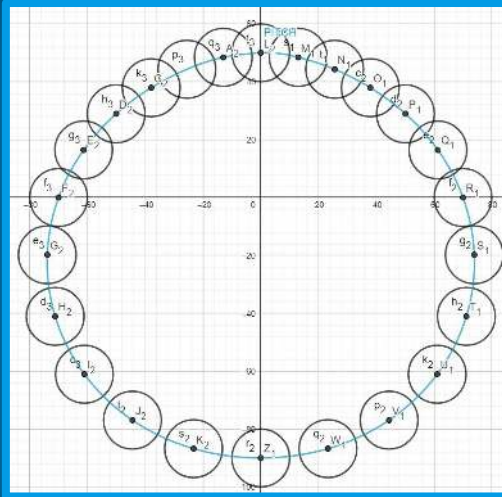
STEP3

We then use the spline function to draw a curve through all the points marked in the previous step, which forms the pitch curve (we are doing this first for convenience).



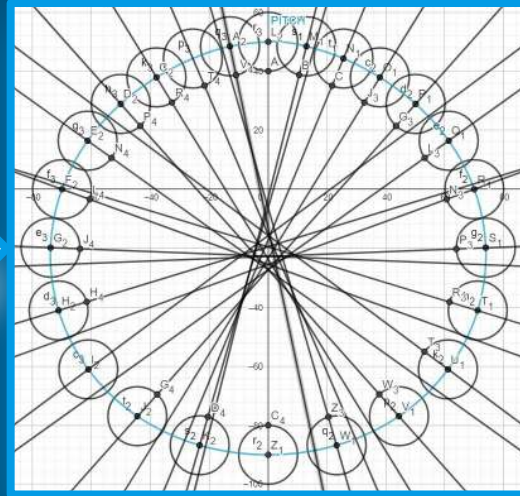
STEP4

After this, we draw circles of roller radius (5 mm) with centres at the points marked in step 2.



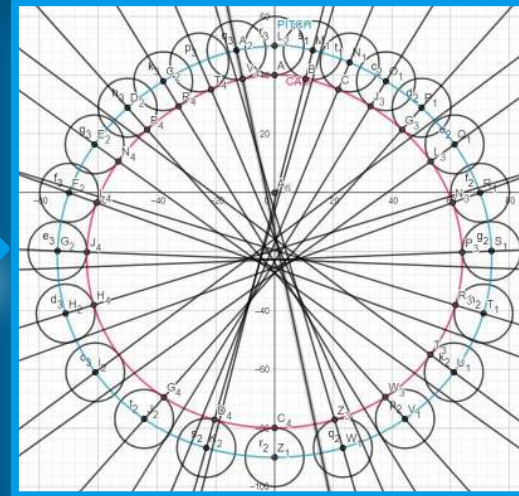
STEP5

We then draw normals to the pitch curve through the centre of each circle (using tangents drawn to pitch curve at those points), and the points where these intersect the respective circle are marked (inner side).

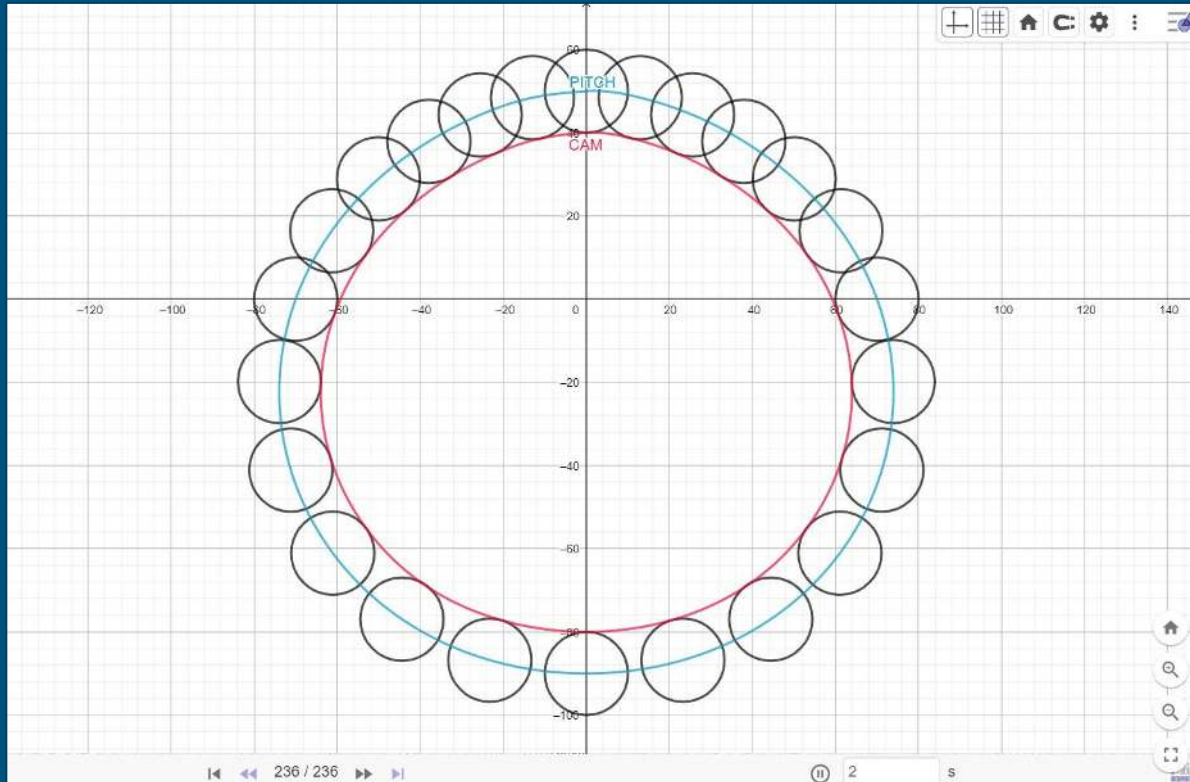


STEP6

We then use the spline function again to plot an *approximate* cam profile for the given specifications, thus completing the construction.



Final profile:



Geogebra file - <https://www.geogebra.org/classic/jtztxsyh>

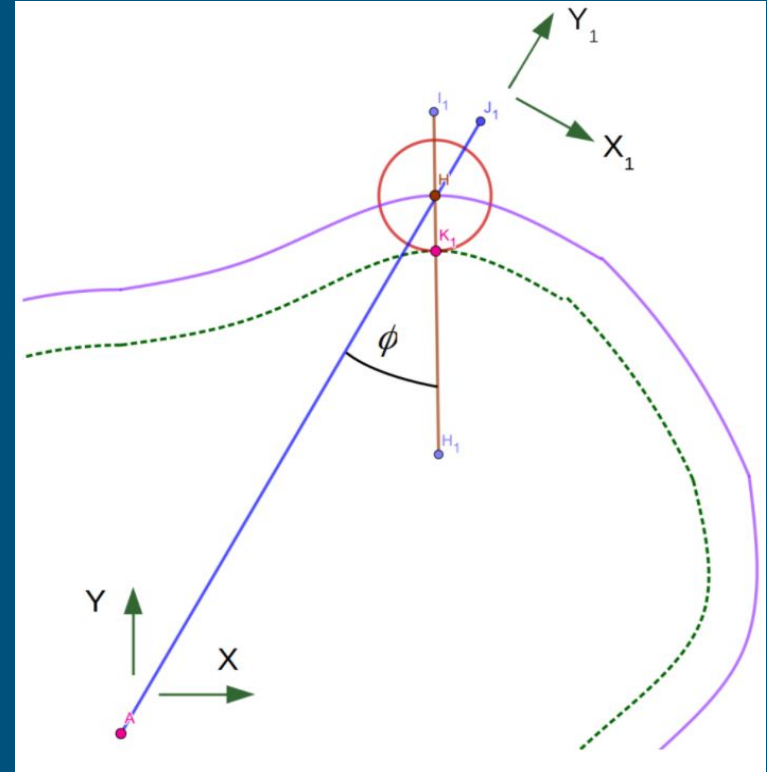
Analytical approach:

- Analytically ,we can synthesise the cam profile by formulating parametric equations X_c and Y_c in terms of θ
- First, we can find the coordinates of Pitch curve by simply taking component of lift in the X and Y axes

$$X_p = (R_p + y(\theta))\sin(\theta)$$

$$Y_p = (R_p + y(\theta))\cos(\theta)$$

- A is the origin, we require the coordinates of the vector $A\vec{K}_1$
- We have coordinates of H in the rotating coordinates as $(0, R_p + y(\theta))$
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- So we need to add $A\vec{H}$ and $H\vec{K}_1$
- is given by $(R_r \sin(\phi), -R_r \cos(\phi))$
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- $A\vec{K}_1$ is given by $A\vec{H} + H\vec{K}_1$
- $(R_r \sin(\phi), R_p + y(\theta) - R_r \cos(\phi))$



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- Now we transform to XY coordinate system:

$$X_c = (R_r \sin(\phi)) \cos(\theta) + (R_p + y(\theta) - R_r \cos(\phi)) \sin(\theta)$$

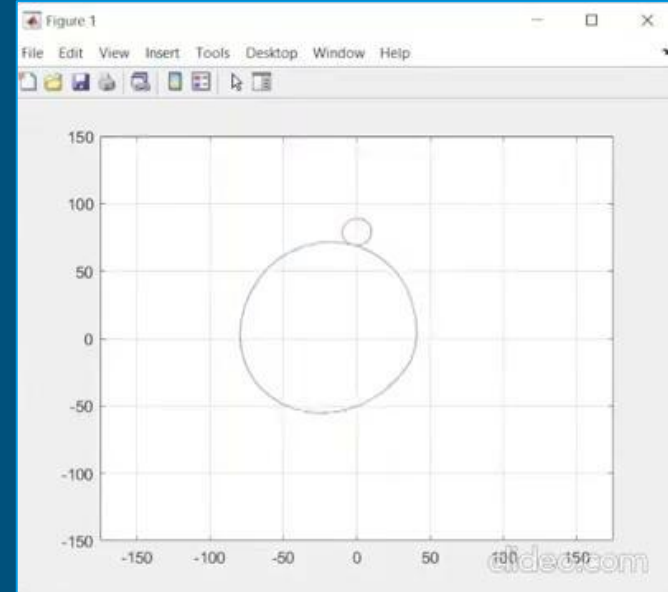
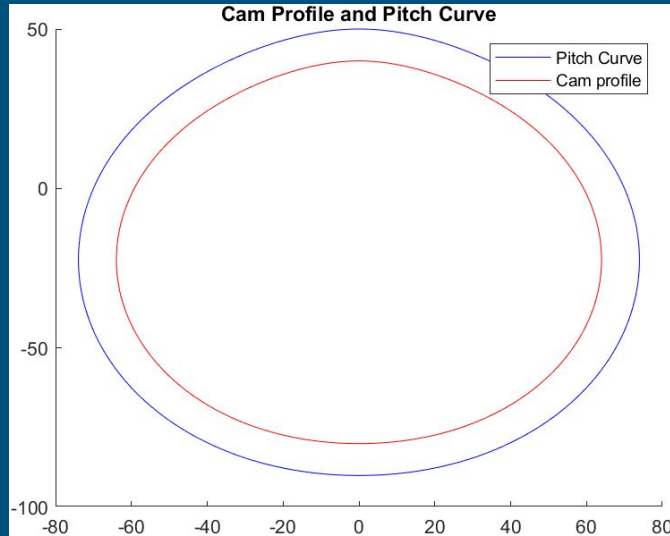
$$Y_c = -(R_r \sin(\phi)) \sin(\theta) + (R_p + y(\theta) - R_r \cos(\phi)) \cos(\theta)$$

- For return, the sign of phi is negative, so the coordinates are given by:

$$X_c = (-R_r \sin(\phi)) \cos(\theta) + (R_p + y(\theta) - R_r \cos(\phi)) \sin(\theta)$$

$$Y_c = (R_r \sin(\phi)) \sin(\theta) + (R_p + y(\theta) - R_r \cos(\phi)) \cos(\theta)$$

- On running the matlab code, we get the pitch curve and the cam profile as follows:



Click on the pic for the video

Matlab code link:

<https://drive.google.com/file/d/1cWymPQFacDsClpy82YExh0nMChGyjlhb/view?usp=sharing>